

RESEARCH INTERESTS

Random matrix theory, high-dimensional probability, AI for mathematics, theoretical computer science, and mathematical foundations of data science.

PUBLICATIONS

- Poisson statistics of one-dimensional random band matrices** 2026
Jiaqi Fan and Guangyi Zou; arXiv:2609.23389; preprint
- Total Variation Distance between Product Distributions: an Analytic Proxy** 2026
Ariel Avital, Aryeh Kontorovich, Roman Vershynin, and Guangyi Zou; arXiv:2609.21049; submitted
- On the Subgaussianity of Quantized Linear Maps: An AI-Assisted Note** 2026
Guangyi Zou and Roman Vershynin; arXiv:2605.27563; ICML 2026 AI4Math Workshop
- Edge Universality for Inhomogeneous Random Matrices II: Markov Chain Comparison and Critical Statistics** 2026
Dang-Zheng Liu and Guangyi Zou; arXiv:2604.20215; submitted
- Edge Universality for Inhomogeneous Random Matrices** 2025
Dang-Zheng Liu and Guangyi Zou; arXiv:2508.17838; submitted
- Revisit the Partial Coloring Method: Prefix Spencer and Sampling** 2025
Dongrun Cai, Xue Chen, Wenxuan Shu, Haoyu Wang, and Guangyi Zou; arXiv:2408.13756; COCOON 2025
- Outliers for Deformed Inhomogeneous Random Matrices** 2024
Ruohan Geng, Dang-Zheng Liu, and Guangyi Zou; arXiv:2407.12182; submitted
- Edge Statistics for Random Band Matrices** 2023
Dang-Zheng Liu and Guangyi Zou; arXiv:2401.00492; submitted

EDUCATION

Ph.D.	University of California, Irvine	Advisor: Prof. Roman Vershynin	Sep 2025 – Present
M.Sc.	University of Science and Technology of China	Advisor: Prof. Dang-Zheng Liu	Sep 2022 – Jun 2024
B.Sc.	University of Science and Technology of China		Sep 2018 – Jun 2022

RESEARCH EXPERIENCE

- Research Assistant, University of Science and Technology of China** Mar 2025 – Jul 2025
Supervisor: Prof. Dang-Zheng Liu
- Research Assistant, Tsinghua University** Sep 2024 – Jan 2025
Supervisor: Prof. Fan Yang
- Academic Visitor, Peking University** Jan 2022 – May 2022
Host: Prof. Xinyi Li

AWARDS

Euler Outstanding Promise Award	2026
Anhui Province Outstanding Graduate (top 4%)	2024
Graduate Scholarship (2×10,000 RMB)	2022, 2023
One Silver Prize and Three Bronze Prizes, S.-T. Yau College Student Mathematics Contest	2020, 2021
Yang Yuanqing Education Foundation Scholarship (top 1%)	2020

LEADERSHIP AND TEACHING

- Teaching Assistant, Probability Theory** Fall 2021; Spring 2022, 2023, 2024
University of Science and Technology of China
- President, Student Algorithm Club** Jun 2020 – Jun 2021
University of Science and Technology of China

TECHNICAL SKILLS

Programming: Python, C++ Typesetting: L^AT_EX